

# ELI MK-XI — Full Project Audit (Engineering Reference)

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**Purpose:** A durable, accurate map of the codebase to be used when **refactoring, upgrading, and fixing**. It records what each subsystem is, where it lives, its entry points, its current status, and every known issue — so future work starts from facts, not guesses.

**Snapshot:** commit a641471, 2026-06-17. Counts marked (*runtime snapshot*) come from the live EXPLAIN\_MEMORY\_RUNTIME dump in the 2026-06-17 session and drift over time.

**Companion docs:** blueprints/adaptive\_inference\_governor.md (deep-dive on the think/budget/throughput policy), plus the existing blueprints/\*.md set (indexed in §13).

### Accuracy policy (how to trust this document)

Every statement is one of: - **[V]** verified this session by reading the code / running it (file:line given); - **[L]** evidenced by the 2026-06-17 session log (quoted/derived); - **[B]** sourced from an existing blueprint or the maintainer memory index, **not** re-verified line-by-line this session (flagged so it can be re-checked before relying on it).

If a statement has no tag it is a structural fact from the file tree / wc / grep run this session **[V]**.

## 1. At a glance (verified this session)

| Metric                  | Value                                                                   |
|-------------------------|-------------------------------------------------------------------------|
| Python files under eli/ | <b>370</b> (+ <i>api/server.py, the FastAPI web app, ~3,884 lines</i> ) |
| Python LOC under eli/   | <b>143,432</b> ( <i>measured 2026-07-01</i> )                           |

| Metric                                   | Value                                                                                                                                                                                                                |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Largest modules                          | executor_enhanced 14,294 · engine 13,326 · GUI<br>eli_pro_audio_gui_MKI 10,985 · router 7,076 ·<br>labs_tab 5,562 · memory 4,415 ·<br>deterministic_grounding_gate 4,292 · agent_bus<br>3,073 · gguf_inference 2,538 |
| Capabilities (capability_manifest.json)  | <b>208</b>                                                                                                                                                                                                           |
| SUPPORTED_ACTIONS (executor_enhanced.py) | <b>174</b>                                                                                                                                                                                                           |
| Specialist bus agents                    | <b>14</b> (named in §4.4)                                                                                                                                                                                            |
| Test files / suite                       | <b>189</b> files; full run <b>7,347 passed / 5 failed / 45<br/>skipped / 2 xfailed (2026-07-01)</b>                                                                                                                  |
| Test groups                              | tests/claims (11), tests/regression (6),<br>tests/generated (3)                                                                                                                                                      |
| SQLite stores                            | <b>4</b> (user, agent, system_index, coding_memory)                                                                                                                                                                  |
| Vector index                             | FAISS, <b>1,228</b> vectors,<br>nomic-embed-text-v1.5.Q4_K_M ( <i>runtime<br/>snapshot</i> )                                                                                                                         |
| Knowledge graph                          | <b>50</b> entities / <b>49</b> relations ( <i>runtime snapshot</i> )                                                                                                                                                 |
| Entry point                              | python -m eli → eli/__main__.py:main();<br>engine CognitiveEngine.process at<br>engine.py:8790                                                                                                                       |

### Subpackage LOC map (eli/)

| Package   | files | LOC    | Role                                                                          |
|-----------|-------|--------|-------------------------------------------------------------------------------|
| execution | 14    | 23,082 | router +<br>executor<br>(the<br>action<br>surface)                            |
| runtime   | 79    | 22,225 | grounding<br>gate,<br>guards,<br>con-<br>tracts,<br>remedia-<br>tion —<br>the |
| gui       | 6     | 17,618 | catch-all<br>PyQt app<br>+ tabs                                               |
| kernel    | 8     | 14,406 | engine.py<br>cogni-<br>tion<br>con-<br>troller +<br>state                     |
| cognition | 27    | 13,739 | agents,<br>infer-<br>ence,<br>persona,<br>reason-<br>ing<br>modes             |

| Package                                  | files | LOC      | Role                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|-------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| memory                                   | 13    | 6,936    | SQLite +<br>FAISS +<br>KG +<br>recall<br>vision,<br>STT, TTS,<br>screen,<br>gaze<br>hardware,<br>settings,<br>paths,<br>model<br>tier, dag,<br>netguard<br>proactive<br>daemon,<br>jobqueue,<br>sched-<br>uler,<br>world-<br>room<br>self-<br>model,<br>habits,<br>sum-<br>maries<br>frontier<br>coding<br>agent<br>(plan→solve→verify)<br>news<br>fetcher,<br>image<br>engine<br>(im-<br>age_engine<br>nested)<br>logging,<br>con-<br>tracts,<br>plugin<br>host,<br>world<br>avatar<br><b>empty/dead</b><br>(see §10) |
| perception                               | 20    | 6,821    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| core                                     | 23    | 6,073    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| planning                                 | 24    | 3,795    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| learning                                 | 12    | 3,315    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| coding                                   | 12    | 2,031    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| tools                                    | 4*    | 1,903    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| utils/contracts/plugins/system/world/cli |       | <1k each |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| guards/integrations                      | 1     | 0        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## 2. Repository map (top level)

- eli/ — the application (above).
- blueprints/ — design docs (index §13). This audit + the governor plan live here.
- artifacts/ — runtime state: db/ (the 4 SQLite stores), vectors/ (FAISS index+meta), runtime\_snapshot.json, runtime\_hardware\_profile.json, analyze\_image\_\*, eval outputs, world snapshots.

- models/ — GGUF weights (text + embeddings + vision + LoRA); no model is baked into code.
  - tests/ — pytest suite (§11).
  - tools/, scripts/, ops/, packaging/, training/ — eval/build/train/ops.
  - config/ — trusted agents, settings.
  - Install: install.sh / install.bat / install.ps1, requirements\*.txt, requirements.lock.txt, pyproject.toml.
- 

### 3. Control flow (the cognition pipeline)

Single entry: `CognitiveEngine.process()` (engine.py:8790). 12 logical stages **[L/V]** (stage map verified live via `EXPLAIN_COGNITION_RUNTIME`):

1. Perceive + ingest — `gui/eli_pro_audio_gui_MKI.py` (text) / `perception/audio_stt.py` (voice).
2. Input normalization + guards — `engine.py` input guards (long-question, command, wake-word).
3. Router + task decomposer — `execution/router_enhanced.py:route()` → {action, args, confidence, meta}.
4. Truth / grounding gate — forbids a direct LLM answer when grounding is required.
5. Executive controller / planner — picks agent profile + order.
6. Agent bus — up to 14 specialist agents on a `ThreadPoolExecutor`, ordered by a dependency DAG.
7. Working-memory / context assembler.
8. **Single inference broker** — `cognition/inference_broker.py`; all model calls serialize on `_LLM_CALL_LOCK` (gguf\_inference.py:515). llama.cpp is not concurrency-safe; vision hot-swap and the ambient daemon coordinate on this lock too. **[V]**
9. Reasoning / synthesis — pure chat / grounded / hybrid; reasoning-mode strategies in §7.
10. Output governor — `cognition/output_governor.py` (1,200 LOC) shape/evidence/confidence.
11. Delivery — GUI / TTS (`perception/tts_router.py`, Piper).
12. Learning + state update — `memory/memory.py` stores turns/memories/habits/failures.

**Fast path:** `_PHASE45_DIRECT_FAST_ACTIONS` (engine.py:78) — deterministic actions whose executor result is authoritative are executed and returned verbatim, bypassing synthesis (e.g. `SCREENSHOT`, `ANALYZE_IMAGE`, `CHECK_JOB`, and now `CREATE_FILE` after a641471). **[V]**

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### 4. Subsystem audit

**4.1 Router** — `execution/router_enhanced.py` (7076) Deterministic, priority-ordered intent parser: explicit regex routes → `multi_commandprepass` (`runtime/command_splitter.py`) → LLM intent-resolver fallback (`cognition/llm_intent.py`). Public `entryroute(text)`. **\*\*Status:\*\*** the highest-churn correctness surface; regex over/under-capture is the dominant bug class (see §9). Fixed in a641471: job-status, LIST\_DIR over-capture, CREATE\_FILE dir capture, multi-command banter. **\*\*Refactor note:\*\*** the file is a long if-ladder; a `route-table` + per-route unit fixtures (`tests/router_test_data.json`)

already exists) would make ordering bugs (like the job-vs-list precedence) visible.

**4.2 Executor** — `execution/executor_enhanced.py` (14340) Implements **\*\*174\*\*SUPPORTED\_ACTIONS(:18** Each action is an if a == "X": block returning {ok, action, content, response, ...}. Security-gated file writes (`CREATE_FILE:8829` via `runtime/security.SecurityManager.is_path_allowed`). Background work offloaded via `runtime/background_tasks.py` (`CHECK_JOB/BACKGROUND_JOBS` at:8679 **\*\*Status:\*\*** largest file in the repo; the action-dispatch is a giant linear scan. **\*\*Refactor note:\*\*** a dispatch dict {ACTION: handler}' would cut the scan and make the **action surface**

enumerable for docs/tests.

**4.3 Kernel / engine** — kernel/engine.py (13374) **CognitiveEngine** owns the pipeline, mode contracts, grounded-control synthesis, and the reasoning-strategy runners (`_run_chain_of_thought:6131`, `_run_tree_of_thoughts:6223`, `_run_self_consistency:6506`). Grounded synthesis helps `_compact_grounded_synthesis(≈12946)` and `_synthesize_answer(13118)`. **\*\*Status:\*\*** second-largest file; very high coupling. The empty-‘cascade originates here (synthesis call shapes) — see governor

doc. **Refactor note:** the synthesis/think/budget logic should move behind the Adaptive Inference Governor (companion doc) rather than per-site flags.

**4.4 Agent bus** — cognition/agent\_bus.py (3138) **\*\*15 agents\*\*** (verified classes): `memory(723)`, `system`. Profiles selected per action by `_select_profile` (default branch + keyword heuristics; this is what dropped `system` on the compound `CREATE_FILE` turn — §9). Dependency DAG: `knowledge_graph ← memory(_AGENT_DEPENDENCIES,:596)`, executed in topological layers via `core/dag.py`. **\*\*[V]\*\*** **\*\*Note [B]:\*\*** memory index records `FrontierAgent` as dormant and bus agents as

retrieval-level rather than frontier-quality — not re-verified this session.

**4.5 Inference** — cognition/gguf\_inference.py (2677) + `inference_broker.py` llama.cpp wrapper. Model-agnostic resolution (`get_model_path`), family-aware templating (`_is_chatml_model/_is_llama_model/_is_mistral_model`, filename-based), no-think prefill (`_no_think_prefill`), thinking-model detection (`_is_thinking_model:219`, filename allowlist), live decode-speed EMA (`record_speed`, non-stream path:1162). All calls serialize on `LLM_CALL_LOCK`. **\*\*Status:\*\*** the think/budget policy is incomplete for cross-hardware use — full analysis + plan in `inadaptive_inference_governor.md`. `force_no_think()` added in `a641471`

(one caller). **[V]**

**4.6 Memory** — memory/memory.py (4461) + `memory/4` SQLite stores + FTS5 mirrors + FAISS + KG (§5). Hybrid recall: SQLite LIKE + FTS5 + FAISS + KG, merged/reranked (`cognition/reranker.py`, `cognition/scoring.py`). Entry classes `Memory`, `get_memory()`, `recall_memory()` **[L/V]**

**Note [B]:** memory index records 3 memory-ranking fusion weight sets that still differ (deferred tuning) — not re-verified.

#### 4.7 Perception — perception/ (20 files)

- **Vision** `vision.py`: model-agnostic VLM via llama.cpp multimodal; mtmd encoder forced to CPU to avoid a CUDA clip segfault (:459). **Documented default Qwen2.5-VL; runtime in the log resolved Moondream2** (`MoondreamChatHandler`) — a discrepancy (§9). OCR path exists (`analyze_image.py`, tesseract) but “read the text” routed to captioning in the log. **[L/V]**
- **STT** `audio_stt.py` (1848): faster-whisper small.en, CPU, int8 (*runtime snapshot*). **[L]**
- **TTS** `tts_router.py`: Piper-only, en\_US-amy-medium, aplay. **[L]**
- Also: screen read/analyze, gaze engine, ambient vision daemon, PDF/CSV analysis.

#### 4.8 Coding agent — coding/ (12 files, 2,031)

Self-contained frontier coding subsystem: plan → DAG/tree search → execute → verify → repair (`coding/__init__.py` docstring **[V]**), sandbox at `coding/sandbox.py`, `plan_graph.py`, `cost.py` (`should_background`). Invoked by `CODE_SOLVE` and by self-improvement. **[V] Note [L]:** the coding agent’s fix-generation calls hit the same empty-‘<think> failure (`SELF_IMPROVE` turn) — covered by the governor plan.

#### 4.9 Planning / proactive — planning/ (24 files)

proactive\_daemon.py (1301): continuous habit/pattern learning, persona-overlay refresh, autonomy tick (code-monitor + self-model + scheduler proposals), news synthesis. Also jobqueue.py (durable subprocess jobs — distinct from in-process runtime/background\_tasks.py), schedulers, and the world-room avatar. **Status [L]:** the daemon emits very frequent persona\_updater/kg sync churn (§9 G). **[V/L]**

**4.10 Learning / self-improvement — learning/ (12) + runtime/self\_improvement.py (1206)** Self-model refresh, session summaries, habit detection/rules, failure analysis (**SELF\_ANALYZE**), fix proposals (**SELF\_IMPROVE**→ coding agent, propose-only). **self\_improvement.py:641**uses **broker.infer(max\_tokens=700)**(small → already no-think). **\*\*[V]\*\*** **\*\*Note [B]:\*\*** memory records **profile\_extractor** as a regex→canned-phrase table with a hard-coded **active\_project="developing ELI"**, and **time\_habit'** as stale/unwindowed — not

re-verified this session.

#### 4.11 Core — core/ (23 files)

Hardware + runtime substrate: hardware\_profile.py (1,040, free-VRAM sizing, compute reserve :69), startupHardware\_optimizer.py (boot profile, train\_ctx\_for\_model:164), model\_tier.py (size tier + **live tok/s EMA**), runtime\_settings.py (1,034, portable storage / path healing), paths.py/portable\_paths.py/legacy\_paths.py/db\_paths.py, netguard.py (204, offline-by-default socket failsafe), dag.py (474, orchestrator), crisis\_guard.py, full\_control.py, model\_download.py, cognition\_tunables.py. **Finding [V]:** dynamic\_runtime\_budget.py (DynamicRuntimeBudget) has **zero references repo-wide** — a dead/unwired module that is a half-built hardware-budget the governor could fold in (§10).

#### 4.12 Runtime — runtime/ (79 files, 22,225)

The catch-all: deterministic\_grounding\_gate.py (4292), grounded\_remediation.py (1604), generated\_script\_guard.py (1072), control\_contracts.py (943), security.py (207), approval\_engine.py (103), background\_tasks.py, command\_splitter.py, evidence/identity/ operator/response sub-families. **Refactor note:** this package is the least cohesive; many single-purpose guard/contract files. A pass to group by concern (grounding / guards / evidence / contracts) would improve navigability.

#### 4.13 GUI — gui/ (6 files, 17,618)

PyQt app eli\_pro\_audio\_gui\_MKI.py (10,985) + labs\_tab.py (5696) + panels. 12 main tabs **[B]** (Report Builder promoted; Proactive×6 / Labs×10 / Settings×5 sub-tabs per memory index). QScintilla optional (falls back to basic editor). **Status:** large monolith GUI files.

#### 4.14 World / plugins / tools / cli

- world/ — avatar “rooms” (Reflection Chamber, Memory Archive); gated to world/activity queries (a greeting leaked a room narration in the log — §9). **[L]**
- plugins/ — plugin host (3 files).
- tools/ — news/news\_fetcher.py (RSS/arXiv/HN/Reddit), image\_engine/ (SSD-1B + procedural).
- cli/ — headless REPL pieces; eli/\_\_main\_\_.py --headless, --trust-agent.

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### 5. Data & state stores (table counts: runtime snapshot 2026-06-17)

Path: artifacts/db/. **[L]** (from EXPLAIN\_MEMORY\_RUNTIME)

| Store                 | Size     | Tables | Role                                                                                                                                                                                                                  |
|-----------------------|----------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| user.sqlite3          | 58.32 MB | 38     | the real store:<br>memories,<br>conversation_turns,<br>kg_entities/relation<br>news,<br>habits,<br>observations,<br>re-call_log,<br>runtime_events,<br>session_summaries,<br>working_memory_pins,<br>+ FTS<br>mirrors |
| agent.sqlite3         | 2.66 MB  | 25     | agent_dispatches,<br>agent_metrics,<br>code_patches,<br>failures,<br>improvements,<br>observations                                                                                                                    |
| system_index.sqlite3  | 0.566 MB | 4      | desktop_apps,<br>executables,<br>recent_files,<br>user_dirs                                                                                                                                                           |
| coding_memory.sqlite3 | 0.02 MB  | 1      | coding_bug_fixes                                                                                                                                                                                                      |

- **FTS5 mirrors:** memories\_fts, kg\_entities\_fts, news\_fts.
- **FAISS:** artifacts/vectors/index.faiss (+meta.json), 1,228 vectors, dim 768, embedder models/embeddings/nomic-embed-text-v1.5.Q4\_K\_M.gguf.
- **KG:** kg\_entities (50) / kg\_relations (49).
- Live counts at snapshot: memories 1,265; conversation\_turns ~4,426; distinct sessions 967; learning\_replay 6,189; runtime\_events 16,426; news\_articles 1,801.
- **Note:** “active/user/memory DB roles alias the same user.sqlite3 file” — 4 *physical* files, more logical roles. **[L]**

## 6. Action / capability surface

- capability\_manifest.json — **208** capabilities (regenerated at startup; generated\_at 2026-06-17). capability\_inventory.generated.json is the generated inventory.
- executor\_enhanced.SUPPORTED\_ACTIONS — **174** action strings (:1899). The gap between 208 capabilities and 174 actions is expected (capabilities include GUI/registry items that aren’t executor actions). **Note:** the maintainer-memory figure “155 actions / 194 caps” is **stale**; the verified current

numbers are 174 / 208.

- Full human catalogue: blueprints/capabilities\_and\_actions.md, capability\_catalogue.md. **[B]**

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## 7. Reasoning modes & inference policy

- Modes (cognition/reasoning\_modes.py): quick, normal, advanced, research, expert (visible names), plus strategy runners CoT / ToT / self-consistency / constitutional. **[V]**
- Per-mode budget (:312–352): derived from n\_ctx, prompt pressure, query complexity, mode — **no throughput term**. Pass-count (not length) is capped by measured tok/s via model\_tier.speed\_passes() (engine :6241, :6520). **[V]**
- **This is the active work area**. The full analysis (think/no-think, throughput-blind budgets, filename-based capability detection, empty-<think> recovery, the unused DynamicRuntimeBudget) and the proposed **Adaptive Inference Governor** are in blueprints/adaptive\_inference\_governor. Treat that as the authority for this layer.

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## 8. Cross-cutting invariants (must hold across all changes) [B, maintainer policy]

1. **Model-agnostic** — no hard-coded model name/size on the inference path; frontier-quality bar.
2. **No fake actions** — if ELI says it did X, the pipeline must actually do X (no narrated success).
3. **Offline-by-default** — all internet via core/netguard; process-wide socket failsafe.
4. **Redistributable / no personal data** — never hard-code a user name or /home/<user> path in source; runtime redaction exists. Settings are stored portably (\_portable\_settings\_for\_storage).
5. **Grounding** — grounded/self-referential questions must use real evidence, never CHAT-guessing; degenerate -/empty fragments are discarded, not surfaced.
6. **Single inference broker / serialized GGUF** — one model call at a time (\_LLM\_CALL\_LOCK).
7. **Check the codebase before proposing** — features usually already exist; frame work as gap-analysis, not greenfield.

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## 9. Consolidated known-issues register

Severity P0=integrity, P1=routing/exec, P2=quality. Status as of a641471.

| #  | Issue                                                                     | Where                     | Status                   |
|----|---------------------------------------------------------------------------|---------------------------|--------------------------|
| R1 | Job-status queries ("job #3") didn't reach CHECK_JOB (# broke regex)      | router                    | <b>FIXED<br/>a641471</b> |
| R2 | CREATE_FILE dropped target dir + wasn't executed (profile dropped system) | router + engine fast-path | <b>FIXED<br/>a641471</b> |
| R3 | LIST_DIR over-capture (swallowed trailing clause; "my notes"→dir)         | router                    | <b>FIXED<br/>a641471</b> |
| R4 | Multi-command banter folded into media title                              | command_splitter          | <b>FIXED<br/>a641471</b> |



| #   | Issue                                                                                           | Where                            | Status                                                                          |
|-----|-------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------------------------------------------------------|
| R5  | EXPLAIN_* compact synthesis<br>empty-<think> loop                                               | engine + gguf                    | <b>PARTIAL (compact) a641471; governor plan for the rest</b><br>OPEN → governor |
| I1  | Catastrophic latency (2-9 min/turn) — throughput-blind budgets on slow HW                       | inference policy                 | OPEN → governor                                                                 |
| I2  | Standard-synthesis fallback + coding fix-gen still empty on <think>                             | engine + coding                  | OPEN → governor                                                                 |
| I3  | Thinking-detection is a filename allowlist                                                      | gguf_inference:219               | OPEN → governor                                                                 |
| I4  | ctx table undershoots model n_ctx_train (32768 vs 262144)                                       | train_ctx_for_model              | OPEN → governor                                                                 |
| I5  | LIST_NOTES success discarded by low-grounding downgrade→CHAT (wrong “just one”)                 | engine                           | OPEN (parked)                                                                   |
| I6  | Confidence gate checks response conf, not grounding (passes grounding=0.0)                      | engine stage 12                  | OPEN                                                                            |
| I7  | Vision = captioning, not OCR; “read the text” didn’t OCR; Moondream2 vs documented Qwen2.5-VL   | perception/vision                | OPEN                                                                            |
| I8  | Memory-count inconsistency (3295 vs grounded 1265) across phrasings                             | memory introspection             | OPEN                                                                            |
| I9  | Multi-part question half-answered (persona Q ignored)                                           | engine/synthesis                 | OPEN                                                                            |
| I10 | World-room narration leaked into a plain greeting                                               | world + engine gate              | OPEN                                                                            |
| I11 | Proactive persona_updater/KG-sync churn (deterministic, redundant; CPU/DB, <b>not</b> LLM-lock) | proactive_daemon/persona_updater | OPEN → governor §G                                                              |

| #       | Issue                                                                                                                           | Where              | Status                       |
|---------|---------------------------------------------------------------------------------------------------------------------------------|--------------------|------------------------------|
| I12 [B] | profile_extractor<br>regex→canned,<br>active_project<br>hard-coded; time_habit<br>stale; frequency-habits<br>never become rules | learning/proactive | OPEN,<br>not re-<br>verified |
| I13 [B] | FrontierAgent<br>dormant; bus agents<br>retrieval-level not<br>frontier                                                         | agent_bus          | OPEN,<br>not re-<br>verified |

## 10. Dead / unwired / duplication (refactor targets)

- **core/dynamic\_runtime\_budget.py** — DynamicRuntimeBudget has **zero references** repo-wide. Dead; overlaps the governor's intended budget object. **[V]**
- **eli/integrations/** (1 file, 0 LOC) — empty package. (eli/guards/ was the other empty package; it has been **removed**.) **[V]**
- **eli/tools/image\_engine** appears both as a 1,750-LOC module and a nested image\_engine/image\_engine/ package (engine 1,483 + visual\_core 1,447) — possible duplication to reconcile. **[V]**
- **Two inference-broker concepts** — verified the live one is cognition/inference\_broker.py; core/inference\_broker.py does **not** exist. Don't reintroduce a second.
- **[B] duplication convergence** (memory): canonical cognition/scoring.py exists; topic-stopwords / FTS-splits / two-tier routing are distinct (not dup); 3 memory-ranking fusion weights still differ — deferred.
- Executor (174-way if) and router (long if-ladder) are the two biggest *structural* refactor candidates (dispatch table + route table).

## 11. Tests & verification

- 194 test files; full run **7,347 passed / 5 failed / 45 skipped / 2 xfailed** (2026-07-01). **[V]** The 5 remaining reds are the in-progress smart\_home plugin removal + one stale blueprint ref — pre-existing, unrelated, fail identically on a clean tree.
- Groups: tests/claims/ (project-vs-claims, symbol inventory, agent contracts), tests/regression/, tests/generated/ (+ \_manifest.json), plus the flat test\_\* suite.
- No-GGUF pattern: many tests assert engine/router behaviour without loading a model (tests/test\_\*\_no\_gguf) — the right place to add governor decision tests.
- Isolation via in-project ELI\_ARTIFACTS\_DIR (no monkeypatch). **[B]**
- Report: tools/run\_test\_report.py → artifacts/test\_report.md; run\_tests.sh.
- **Baseline at a641471:** router/multi-command/background (239) + filesystem/think-stop/reasoning/runtime-status (61) green; full collection clean. **[V]**

## 12. Install / packaging / redistribution

- install.sh (+ .bat/.ps1): frozen lock (requirements.lock.txt) + GPU verify + DB init. **[B]**
- Platform requirement files: requirements-{windows,macos,android,full,learning}.txt.
- pyproject.toml, eli-mkxi.desktop, build\_packages.sh, packaging/.
- Redistribution mechanics: portable settings storage + stale-path healing (runtime\_settings.\_heal\_model\_paths, \_resolve\_relative\_model\_paths), dataset redaction. **Open risk [B]:** personal values (e.g. user\_name) can still end up tracked; the cross-platform generalization mandate (Linux/macOS/Windows) is ongoing.

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### 13. Blueprint index (existing design docs in blueprints/)

architecture.md, architecture\_ascii.md, agent\_algorithms.md, orchestration\_and\_agents.md, dag.md/dag\_orchestrator.md, memory.md, grounding\_and\_evidence.md, inference\_and\_hardware.md, coding\_agent.md, code\_mode\_execution\_layer.md, learning.md, perception.md, gui.md, security.md, operations.md, installation.md, capabilities\_and\_actions.md/capability\_catalogue.md, what\_eli\_can\_do.md/what\_eli\_is.md, project\_overview.md, state\_snapshot.md, runtime\_planning\_world.md, eval\_harness.md, lora\_pipeline.md, background\_tasks.md, model\_bakeoff\_dossier.md, top2\_head2head.md, decomposition\_plan.md, proposal\_total\_awareness.md, complete\_findings.md, diagrams.md. + **this audit** and **adaptive\_inference\_governor.md**.

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### 14. Appendix — verification provenance

**Read this session [V]:** router (route, job/list/create\_file routes, \_expand\_common\_dir), command\_splitter, engine(\_PHASE45\_DIRECT\_FAST\_ACTIONS, \_run\_chain\_of\_thought, \_compact\_grounded\_synthesize\_answer, replan, low-grounding downgrade), gguf\_inference (\_no\_think\_prefill, \_is\_thinking\_model, record\_speed, broker legacy paths), reasoning\_modes budget, model\_tier (EMA + speed\_passes), hardware\_profile(reserve), startup\_hardware\_optimizer.train\_ctx\_for\_model, agent\_bus (15 agents + \_select\_profile branches), persona\_updater, dynamic\_runtime\_budget (dead), executor CREATE\_FILE / CHECK\_JOB handlers, SUPPORTED\_ACTIONS, structure/LOC/test counts. **From the 2026-06-17 session log [L]:** model load, stage map, DB table dump, vision/STT/TTS runtime, latency/empty-<think> timings. **From blueprints / maintainer memory [B], not re-verified:** GUI tab inventory, install internals, FrontierAgent dormancy, profile\_extractor/time\_habit specifics, memory-fusion weights, duplication-convergence status. Re-verify any **[B]** item before relying on it for a fix.